

1. Pipeline processing uses the technique:
 - A. Parallel processing
 - B. Bit slicing
 - C. Sharing the memory
 - D. Prefetching
2. Stack addressing can be classified as:
 - A. Immediate addressing mode
 - B. Implied addressing mode
 - C. Displacement addressing mode
 - D. Direct addressing mode
3. Saturated logic circuits have inherently:
 - A. Low noise immunity
 - B. Low switching speed
 - C. Higher power dissipation
 - D. Short saturation delay time
4. According to Idempotent law, $X+X$ is equivalent to:
 - A. $X-X$
 - B. 0
 - C. 1
 - D. X
5. The octal equivalent of 111010 is:
 - A. 73
 - B. 71
 - C. 70
 - D. 72
6. Which type of memory consists of internal latches that store the binary information?
 - A. An EPROM
 - B. SRAM
 - C. DRAM
 - D. None of above
7. PLA differs from PROM in that:
 - A. PROM provides full decoding of the variable but doesn't generate all the min terms
 - B. PROM doesn't provide full decoding of the variable and doesn't generate all the min terms
 - C. PLA doesn't provide full decoding of the variable and doesn't generate all the min terms
 - D. PLA provides full decoding of the variable but doesn't generate all the min terms.
8. Which one is not a software cost estimation model?
 - A. Computational
 - B. FP (Function point)
 - C. LOC (line of code)
 - D. Empirical
9. Which life cycle is risk driven?
 - A. RAD
 - B. Evolutionary
 - C. Waterfall
 - D. Spiral
10. A programmer would most likely prepare a:
 - A. General system flowchart
 - B. Specific system flowchart
 - C. General program flowchart
 - D. Specific program flowchart
11. In case of software design, functional independence is achieved by:
 - A. Developing least couple modules
 - B. Both A and B above
 - C. Developing highly cohesive modules
 - D. None of the above
12. A software engineering task that bridges the gap between system level requirements engineering and software design is called:
 - A. System design
 - B. System engineering
 - C. Requirements analysis
 - D. None of the above

13. An important aspect in coding is:
 - A. Brevity
 - B. Productivity
 - C. Readability
 - D. To use as small memory space as possible
14. Data flow diagram is a:
 - A. System flowchart
 - B. Data mode
 - C. Process model
 - D. All of the above
15. A smaller specialized version of a data warehouse is called a:
 - A. Data mine
 - B. Data store
 - C. Data element
 - D. Data mart
16. In a relational schema, each tuple is divided into field called:
 - A. Queries
 - B. Domains
 - C. Relations
 - D. None of above
17. The language used for application programs to request data from the DBMS is referred to as the:
 - A. Query language
 - B. DDL
 - C. DML
 - D. None of the above
18. Which of the following field in a student's file can be used as a primary key?
 - A. Roll number
 - B. Name
 - C. Class
 - D. GPA
19. Query optimization mainly relies on the data:
 - A. "Increase the sizes of operand of the unary operations as far as possible"
 - B. "Reduce the sizes of operand of the binary operations as far as possible"
 - C. "Increase the sizes of operand of the binary operations as far as possible"
 - D. "Apply binary operations first if possible"
20. An example of multi-valued attributes would be:
 - A. Phone_number
 - B. Roll_no
 - C. Customer_id
 - D. None of the above
21. A high paging rate:
 - A. Always creates a slow system
 - B. Keeps the system running well
 - C. May cause high I/O rate
 - D. Is a symptom of too much processor activity
22. The operating system manages:
 - A. Disks and I/O devices
 - B. Processor
 - C. Memory
 - D. All of the above
23. A distributed system is the one that runs on a collection of machines that:
 - A. Have shared memory but don't operate in distributed environment
 - B. Do not have shared memory yet looks to its users like a single computer
 - C. Have shared memory and operate in a distributed environment
 - D. Do not have shared memory and looks to its users like a chain of computers
24. The rapid switching of the CPU back and forth from process to process is called:
 - A. Multiprocessing
 - B. Multiprogramming
 - C. Multitasking
 - D. None of the above

25. Which of the following statement concerning storage device is false?
- Main memory is volatile
 - In handling a cache miss, a block (rather than a single word) is transformed from the next level in memory hierarchy.
 - Management of cache memory is responsible to operating system
 - Cache memory is implemented in main memory
26. Which one of the following is the Data Link Protocol?
- Bridges
 - Ethernet
 - Token ring
 - All of the above
27. The difference between IPX protocol and IP protocol can be:
- No difference between in functioning. The only difference is IPX is network protocol whereas IP is an TCP/IP protocol
 - IPX is reliable connectionless internetwork protocol where IP is an TCP/IP protocol
 - Network layer runs on IPX whereas transport layer runs on IP
 - IPX packet is having 10-byte address whereas IP packet has 4-byte address
28. Synchronous protocols:
- Are generally used by personal computers
 - Transmit characters one at a time
 - Allow faster transmission than asynchronous protocols do
 - All of the above
29. Identify the odd term among the following group:
- Microwaves
 - Optical fiber
 - Coaxial cable
 - Twisted pair wire
30. A set of rules governing the format and meaning of the packets or messages that are exchanged by the peer entities within a layer is called....
- A policy
 - A protocol
 - A service
 - None of above
31. Bridges operate in:
- Transport layer
 - Data link layer
 - Physical layer
 - Network layer
32. Returning references to non-const private data:
- Results in a compiler error
 - Allows private functions to be modified
 - Is only dangerous if the binary scope resolution operator (: :) is used in the function prototype
 - Allows private member variables to be modified, thus breaking encapsulation
33. Virtual functions in C++, allows you to:
- Create an array of type pointer to base class that can hold pointers to derived class
 - Use the same function call to execute member function of objects from different class
 - Create functions that have no body
 - Group objects of different classes so they can all be accessed by the same function code
34. If `int S [5]` is a one-dimensional array of integers which of the following refers to the third element in the array?
- `S-2`
 - `*(S+3)`
 - `*(S+2)`
 - `S+3`

35. The inheritance is the sharing of:
- Object
 - Functions
 - Attributes and operations among classes
 - None of the above
36. A base class will offer:
- Be a generalized version of its derived classes
 - Model more specific objects than its derived classes
 - Corresponds to something in the real-world
 - Behave badly when the chips are down
37. Overloaded functions:
- May fail unexpectedly due to stress
 - Are a group of functions, with the same value
 - All have the same number and types of arguments
 - Make life simpler for programmers
38. The time complexity for greedy search algorithm is:
- $O(b^{m/2})$
 - $O(b \times m)$
 - $O(b^m)$
 - $O(m^b)$
39. Lexical part in knowledge representation is associated with:
- Meaning
 - Access procedure
 - Vocabulary
 - Constraints
40. The knowledge base of an expert system includes both facts and:
- Analysis
 - Algorithms
 - Theories
 - Heuristics
41. Which one of the following properties is not expected from good hashing technique?
- Produce no collisions
 - Produce keys uniformly distributed over the range
 - It is easier to program
 - It may induce less collisions
42. The phenomenon where two keys that hash into different values compete with each other in successive rehashes is called:
- Quadratic rehash
 - Primary clustering
 - Recursive clustering
 - None of the above
43. What is the time complexity of linear search algorithm over an array of n elements?
- $O(n^2)$
 - $O(n)$
 - $O(\log_2 n)$
 - $O(n \log_2 n)$
44. Which of the following is least efficient sort?
- Merge sort
 - Quick sort
 - Selection sort
 - Bubble sort
45. Which data structure is needed to convert in fix notations to post fix notations?
- Graph
 - Queue
 - Stack
 - Tree
46. A tree in which the height of the two sub trees of every node never differ by more than one is called:
- Balanced Binary Tree
 - AVL Tree
 - B-Tree
 - A and C of the above

47. In which mapping technique in cache, Replacement Algorithms are not used?
- Set-associative mapping
 - Associative mapping
 - Direct mapping
 - None of the above
48. Which of the following are registers?
- Program counter
 - Stack pointer
 - Accumulator
 - All of the above
49. Which of the following register is loaded with the content of the memory locations pointed by the PC?
- Program counter
 - Memory Data Register
 - Memory Address Register
 - Instruction Register
50. A shift register is particularly useful for:
- Certain hardware multipliers and divides which use a shift add or shift subtract methodology.
 - Certain hardware address and subtractors which use a shift-add or shift subtract methodology
 - Both of the above
 - None of the above
51. An ergodic process is:
- Need not be stationary
 - Always stationary
 - Stationary but not random
 - None of the above
52. A modulator is the system to:
- Extract information from the carrier
 - Amplify the audio frequency signal
 - Impress the information on to a radio frequency carrier
 - Separate two frequencies
53. The adverse effect of noise in a communication system is maximum in:
- Channel
 - Source
 - Encoder
 - Receiver
54. What is the signal velocity if the propagation factor is 0.92 of C?
- 4.02×10^7 m/s
 - 2.76×10^8 m/s
 - 3.24×10^8 m/s
 - 3.45×10^8 m/s
55. What will be the hamming distance between the codes 0101 and 0011 bits?
- 5
 - 3
 - 4
 - 2
56. Which of the following are true of AM/FM system?
- AM/FM is a specialized GIS
 - AM/FM systems are always confined to the utility industry
 - The terms GIS and AM/FM are synonymous
 - AM/FM are always vector based
57. Blackboard pattern is a kind of....
- Software engineering
 - Software testing
 - Software architecture
 - Software project management
58. The study of a current business and information system application and the system is part of which phase?
- Systems analysis
 - Systems planning
 - Systems implementation
 - System design
59. The ...is responsible for maintaining the standards of XHTML on the internet:
- ISO
 - IEEE
 - W3C
 - None of above

60. A logical schema:
- Is a standard way of organizing information into accessible parts
 - Describes how data is actually stored on disk
 - Is the entire database
 - All of the above
61. A computer with 32 k physical memory and a virtual memory generates 16-bit addresses. How many pages are there in the virtual memory if page size is 4k.....
- 2^{16}
 - 16
 - 8
 - None of the above
62. Consider a DFA over $\Sigma = \{a,b\}$ accepting all strings which have number of a's divisible by 6 number of b's divisible by 8. What is the minimum number of states that the DFA will have?
- 15
 - 8
 - 48
 - 14
63. Which of the following statements is false?
- Given the regular grammar, pumping lemma ensures that it is regular
 - Nested IF-ELSE can be represented by CFG
 - S-grammar resembles with GNF as GNF imposes the pair (LHS variable, RHS token) condition
 - For every regular grammar, there exist equivalent DFA, PDA and TURING machine
64. Measurement of algorithmic requires:
- An infinite number of algorithms requires an infinite number of scales in the worst
 - Attributes such as worst – best performance and correctness
 - Sample inputs
 - None of the above
65. Which string can be generated by $S \rightarrow aS/bA, A \rightarrow d/cA$
- abcca
 - Aabccd
 - abababd
 - Adabcca
66. Which statement is true?
- LL (k) parsers read from left to right look at most K token ahead, and perform left most reduction
 - The symbol table stores information about the expressions occurring in a program
 - Left recursive productions naturally lead to right associative parse trees
 - None of the above
67. Lexical analysis in compilation provides except:
- Intermediate code generation augmentation
 - Compiler portability enhancement
 - Compiler efficiency improvement
 - Syntax analysis simplification
68. When is peep hole optimization done?
- Code optimization
 - Lexical analysis
 - Code generation
 - Semantic analysis

69. A language L allows declaration of arrays whose sizes are not known during compilation. It is required to make efficient use of memory. Which one of the following is true?
- A compiler cannot be written for L; an interpreter must be used
 - A compiler using dynamic memory allocation can be written for L.
 - A compiler using static memory allocation can be written for L
 - None of the above
70. Shift reduce or reduce/reduce conflict may arise in which of these:
- Top down bottom up
 - Bottom up
 - Both (A) & (B)
 - Predictive parser
71. Which of these following is not applications graphics software?
- Painting program
 - CAD packages
 - PHIGS
 - Graphing package
72. A common device for drawing, painting, or interactively selecting coordinate position on an object is:
- Digitizer
 - Data glove
 - Scanner
 - Joy stick
73. Moiré or the wavy patterns could be introduced in an image while doing:
- Digitizing
 - Scaling/resizing
 - Filtering
 - Image compression
74. A transistor is a:
- Voltage operated device
 - Both of the above
 - Current operated device
 - None of the above
75. The emitter resistance R_e is bypassed by capacitor to:
- Increase gain
 - Reduce gain
 - Stabilize the Q-point
 - Cause the thermal run-away
76. If biasing is not done in an amplifier, it results in:
- Excessive collector bias
 - Decrease in base current
 - Undefined
 - Unfaithful amplification
77. The impulse response of a matched filter is:
- Exponential function
 - Quadratic function
 - Replica of input signal
 - Time-delayed replica of input signal
78. Four equal charges are located at the corners of a square of side d . if the length of each side of the square is doubled, then what happens to the resultant force on each charge?
- It stays the same
 - It is doubled
 - It decreases by a factor of 4
 - It is quadrupled
79. Consider a 230 v, 100w incandescent lamp. Determine the lamp resistance...
- 530 Ω
 - 529 Ω
 - 524 Ω
 - 528 Ω
80. Which of the following transistor amplifier configuration has the highest power gain?
- Common collector
 - Common base
 - Common emitter
 - None of the above

ANSWER SHEET

1.D	2.B	3.D	4.D	5.D	6.B	7.C	8.A	9.D	10.D
11.C	12.C	13.C	14.C	15.A	16.B	17.C	18.A	19.A	20.A
21.C	22.D	23.B	24.B	25.B	26.B	27.A	28.C	29.A	30.B
31.B	32.D	33.B	34.C	35.A	36.A	37.D	38.C	39.C	40.D
41.A	42.B	43.B	44.D	45.C	46.B	47.D	48.D	49.D	50.C
51.B	52.C	53.A	54.B	55.D	56.A	57.C	58.A	59.C	60.A
61.C	62.C	63.C	64.B	65.B	66.A	67.A	68.C	69.B	70.C
71.A	72.A	73.D	74.B	75.A	76.D	77.D	78.A	79.B	80.C

EXPLANATION OF SOME QUESTIONS:

Question 54*:

The signal velocity (v) can be calculated using the propagation factor (p) and the speed of light in a vacuum (C) using the formula:

$$v = p \cdot C$$

Given that the propagation factor is 0.92 of C , we have:

$$p = 0.92$$

And C is approximately 3×10^8 meters per second.

Substituting these values into the formula, we get:

$$v = 0.92 \times 3 \times 10^8 \sim 2.76 \times 10^8 \text{ m/s}$$

So, the signal velocity is approximately 2.76×10^8 meters per second.

The correct option is:

B. $2.76 \times 10^8 \text{ m/s}$

Questions 55*:

To calculate the Hamming distance between the codes 0101 and 0011, we align them:

0101

0011

The Hamming distance is the number of positions at which the corresponding bits are different.

In this case, the bits at positions 1 and 2 are different. Therefore, the Hamming distance is 2.

So, the correct answer is: D. 2

Questions 61*:

The page size is 4k, which means each page can hold 4 kilobytes of data. Since the computer has 32 kilobytes (32k) of physical memory, and the page size is 4 kilobytes, we can calculate the number of pages in the virtual memory as follows:

$$\text{Number of pages} = \text{Total physical memory} / \text{Page size} = 32\text{k} / 4\text{k} = 8$$

So, there are **8 pages** in the virtual memory.

The correct option is:

C. 8

Question No 79*:

To determine the resistance of the lamp, we can use the formula:

$$P = V^2 / R$$

Given:

- Voltage (V) = 230 V
- Power (P) = 100 W

We can rearrange the formula to solve for resistance (R):

$$P = V^2 / R$$

Substituting the given values:

$$R = (230)^2 / 100 = 529 \Omega$$

$$R = 529 \Omega$$

So, the resistance of the lamp is approximately 529 Ω .

The closest option is:

B. 529 Ω